

is needed to understand the scope and limitations of feature-based steering across different sequence contexts and more complex biological patterns.

4 Discussion

Our work demonstrates that sparse autoencoders can extract thousands of interpretable features from protein language model representations. Through quantitative evaluation against Swiss-Prot annotations, we showed that these features capture meaningful biological patterns that exist in superposition within the model’s neurons. Our analysis framework combines automated interpretation through large language models with biological database annotations, enabling scalable feature characterization. Beyond identifying features that activate on annotated patterns, we showed that feature activation patterns can identify missing database annotations and enable targeted control of model predictions through interpretable steering.

4.1 Implications for PLM Interpretability

The dramatic difference between neuron and SAE feature interpretability (46 vs 2,548 features per layer interpreted with Swiss-Prot concepts) provides strong evidence for information storage in superposition within PLMs. Interestingly, we found that SAEs trained on randomized PLMs still extract features specific to individual amino acids but fail to capture any complex biological concepts we tested for, suggesting that meaningful feature extraction requires learned model weights. This phenomena of identifying amino acids from randomized embeddings aligns with recent observations on randomized language models [20] showing that SAEs capture both properties of the model and the underlying data distribution.

By identifying feature-concept associations through domain-level F1 scores, our biological concept evaluation framework provides a new quantitative approach to assessing interpretability methods, that handles concepts which are annotated more coarsely than the learned features. Furthermore, our LLM-generated feature descriptions achieve strong predictive power at identifying highly activated proteins, even when there are no Swiss-Prot annotations, offering a complementary approach to Swiss-Prot annotations for understanding protein features. While these metrics are only approximations of the true biological interpretability of a model, these metrics clearly distinguish between different approaches and could enable systematic comparison of interpretability techniques. The discovery that "false positive" predictions often indicate missing annotations demonstrates how interpretability tools can provide immediate practical value.

Analysis of feature activation patterns and dictionary vectors reveals both capabilities and open questions about model representations. While we identify many known conserved motifs and 3D patterns, the proportion of features dedicated to motif memorization versus other computational strategies remains unclear. Our framework provides a potential method for quantifying this balance, extending recent work showing PLMs primarily learn coevolutionary statistics [3]. Additionally, our steering experiments demonstrate that features can influence both local and neighboring amino acid predictions, though the precise mechanisms by which features contribute to masked token prediction require further investigation.

4.2 Applications of Interpretable PLM Features

Our SAE framework offers valuable applications across model development, biological discovery, and protein engineering. For model development, these features enable systematic comparison of learning patterns across different PLMs, revealing how architectural choices influence biological concept capture. Feature tracking during training and fine-tuning could provide insights into knowledge acquisition, while analysis of feature activation during failure modes could highlight systematic biases and guide improvements.

Beyond confirming known patterns, these interpretable features may reveal novel biological insights. Features that don’t align with current annotations but show consistent activation patterns across protein families could point to unrecognized biological motifs or relationships that have escaped traditional analysis methods. Additionally, we have demonstrated that feature-based steering can be used to influence sequence generation in targeted ways. While periodic glycine steering may not revolutionize protein engineering itself, it demonstrates a promising new direction for controlling sequence generation during protein design tasks.

4.3 Limitations and Future Directions

Scaling this approach to structure-prediction models like ESMFold or AlphaFold represents a crucial next step. Understanding how learned features evolve with increased model size and structural capabilities could enable fine-

grained control over generated protein conformations through targeted steering. While we demonstrated successful steering for periodic glycine features in fairly simple contexts, extending steering to more complex biological concepts remains challenging, as we need both robust methods and clear success metrics. While structure prediction models could simplify quantitative evaluation of steering outcomes compared to purely sequence-based models, validating that we’ve achieved intended biological changes without disrupting other properties is substantially more complex than evaluating semantic shifts in language models.

Feature interpretability could be improved through both recent advances in SAE training methods and additional evaluation metrics beyond Swiss-Prot annotations. Additionally, our analysis currently focuses only on representations of unmasked amino acids, leaving open questions about the information encoded in masked token embeddings and the CLS token, which has proven valuable as a protein-level representation.

While SAE features reveal learned patterns, what remains unexplored is mapping how these features combine into interpretable circuits for specific capabilities – like 3D contact prediction or binding site and allosteric site identification. Such analysis could illuminate how models extract structural and functional information from individual sequences. Finally, systematic analysis of currently uncharacterized features, guided by their activation patterns and impact on predictions, could reveal novel biological motifs – demonstrating how interpretability tools might contribute directly to biological discovery.

5 Methods

5.1 Sparse Autoencoder Training

5.1.1 Dataset Preparation

We selected 5M random protein sequences from UniRef50, part of the training dataset for ESM-2. For each protein, we extracted hidden representations from ESM-2-8M-UR50D after transformer block layers 1 through 6, excluding <cls> and <eos> tokens. The dataset was sharded into groups of 1,000 proteins each, with tokens shuffled within these groups to ensure random sampling during training.

5.1.2 Architecture and Training Parameters

We trained sparse autoencoders following the architecture described in [14], using an expansion factor of 32x, creating feature dictionaries of size 10,240 from ESM2-8M embedding vectors of size 320. For each layer, we trained 20 SAEs for 500,000 steps using a batch size of 2,048. Learning rates were sampled in increments of 10x from 1e-4 to 1e-8, with L1 penalties ranging from 0.07 to 0.2. Both the L1 penalty and learning rate were linearly increased from 0 to their final values during the initial training phase, with the learning rate reaching its maximum within the first 5% of training steps.

5.1.3 Feature Normalization

To standardize feature comparisons, we normalized activation values using a scan across 50,000 proteins from Swiss-Prot. For each feature, we identified the maximum activation value across this dataset and used it to scale the encoder weights and decoder weights reciprocally, ensuring all features were scaled between 0 and 1 while preserving the final reconstruction values.

5.2 Swiss-Prot Concept Evaluation Pipeline

5.2.1 Dataset Construction

From the reviewed subset of UniprotKB (Swiss-Prot), we randomly sampled 50,000 proteins with lengths under 1,024 amino acids. We converted all binary and categorical protein-level annotations into binary amino acid-level annotations, maintaining domain-level relationships for multi-amino-acid annotations. The dataset was split equally into validation and test sets of 25,000 proteins each. We retained only concepts present in either more than 10 unique domains or more than 1,500 amino acids within the validation set.

5.2.2 Feature-Concept Association Analysis

For each normalized feature, we created binary feature-on/feature-off labels using activation thresholds of 0, 0.15, 0.5, 0.6, and 0.8. We evaluated feature-concept associations using modified precision and recall metrics:

$$\text{precision} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalsePositives}} \quad (1)$$

$$\text{recall} = \frac{\text{DomainsWithTruePositive}}{\text{TotalDomains}} \quad (2)$$

$$\text{F1} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \quad (3)$$

For each feature-concept pair, we selected the threshold yielding the highest F1 score for final evaluation.

5.2.3 Model Selection and Evaluation

We conducted initial evaluations on 20% of the validation set to compare hyperparameter configurations (only including the 135 concepts that had > 10 domains or 1,500 amino acids in this subset alone). For each concept, we identified the feature with the highest F1 score and used the average of these top scores to select the best model per layer. These six models (one per ESM2-8M layer) were used for subsequent analyses and the InterPLM dashboard.

To calculate the test metrics we (1) identify the feature with highest F1 score per-concept on the full validation set, then for each concept, calculate the F1 score of the selected feature on the test set and report these values then (2) identify all feature-concept pairs with F1 > 0.5 in the validation set, calculate their F1 scores on the test set, and report the number of these that have F1 > 0.5 in the test set.

5.2.4 Baselines

To train the randomized baseline models, we shuffled the values within each weight and bias of ESM2-8M, calculated embeddings on the same datasets, and repeated the same training (using 6 hyperparameter choices per layer), concept-based model selection, and metric calculation processes. To compare with neurons, we scaled all neuron values between 0 and 1 (based on the minimum and maximum values found in our Swiss-Prot subset), then input these into an SAE with expansion factor of 1x that has an identity matrix for the encoder and decoder, such that all other analysis can be performed identically.

5.3 LLM Feature Annotation Pipeline

5.3.1 Example Selection

Analysis was performed on a random selection of 1,200 (10%) features. For each feature, representative proteins were selected by scanning 50,000 Swiss-Prot proteins to find those with maximum activation levels in distinct ranges. Activation levels were quantized into bins of 0.1 (0-0.1, 0.1-0.2, ..., 0.9-1.0), with two proteins selected per bin that achieved their peak activation in that range, except for the highest bin (0.9-1.0) which received 10 proteins. Additionally, 10 random proteins with zero activation were included to provide negative examples. For features where fewer than 20 proteins reached peak activation in the highest range (0.9-1.0), additional examples were sampled from the second-highest range (0.8-0.9) to achieve a total of 24 proteins between these two bins, split evenly between training and evaluation sets. Features were excluded if fewer than 20 proteins could be found reaching peak activation across the top three activation ranges combined.

5.3.2 Description Generation and Validation

For each feature, we compiled a table containing protein metadata, quantized maximum activation values, indices of activated amino acids, and amino acid identities at these positions. Using this data, we prompted Claude-3.5 Sonnet (new) to generate both a detailed description of the feature and a one-sentence summary that could guide activation level prediction for new proteins.

To validate these descriptions, we provided Claude with an independent set of proteins (matched for size and activation distribution) along with their metadata, but without activation information. Claude’s predicted activation levels were compared to measured values using Pearson correlation.